Why? Because $|\vec{X}+\vec{Y}|^2$ is just $(\vec{X}+\vec{Y}) \cdot (\vec{X}+\vec{Y})$. Collecting these results, we get

$$|\vec{X}|^2 + |\vec{Y}|^2 + 2|\vec{X}||\vec{Y}| \geq |\vec{X}|^2 + |\vec{Y}|^2 + 2(\vec{X}$$

Now, we just subtract $|\vec{X}|^2 + |\vec{Y}|^2$ from each side and then divide by 2, leaving us with

$$|\vec{X}||\vec{Y}| \geq \vec{X} \cdot \vec{Y}. \qquad (5.6)$$

This is another form of the triangle inequality. It says that, given any two vectors $\vec{X}$ and $\vec{Y}$, the product of their lengths is greater than or equal to their dot product. This is no surprise—the dot product is often defined as

$$\vec{X} \cdot \vec{Y} = |\vec{X}||\vec{Y}| \cos\theta,$$

where θ is the angle between the two vectors. But we know that the cosine of an angle always stays in the range −1 to +1, so the right-hand side must always be less than or equal to $|\vec{X}||\vec{Y}|$. This relationship is true for vectors in two dimensions, three dimensions, or an arbitrary number of dimensions. It's even true for vectors in complex vector spaces. It's generally true for vectors in *any* vector space, provided the length of the vector is defined as the square root of the vector's inner product with itself. As we go forward, we plan to use Inequality 5.6 in the squared form, that is,

$$|\vec{X}|^2|\vec{Y}|^2 \geq (\vec{X} \cdot \vec{Y})^2$$

or

$$|\vec{X}|^2|\vec{Y}|^2 \geq |\vec{X} \cdot \vec{Y}|^2. \qquad (5.7)$$

In this form, it's called the *Cauchy-Schwarz* inequality.